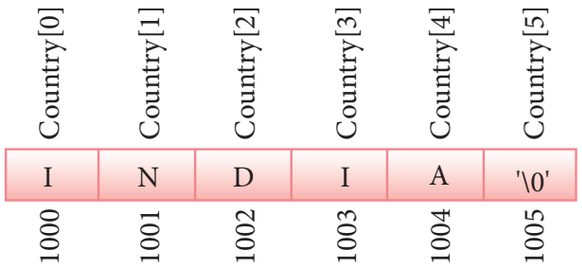

Figure 12.3

In the above memory representation, each character occupies one byte in memory. At the end of the string, a null character is automatically added by the compiler. **C++ also provides other ways of initializing the character array:**

char country[6]={'I', 'N', 'D', 'I', 'A', '\0'};

char country[]="INDIA";

char country[]={'I', 'N', 'D', 'I', 'A', '\0'};

If the size of the array is not explicitly mentioned, the compiler automatically calculate the size of the array based on the number of elements in the list and allocates space accordingly.

In the initialization of the string, if all the characters are not initialized, then the rest of the characters will be filled with NULL.

Example:

char str[5]={'5','+','A'};

str[0]; ---> 5

str[1]; ---> +

str[2]; ---> A

str[3]; ---> NULL

str[4]; ---> NULL

Note

During initialization, the array of elements cannot be initialized more than its size.

For example

char str[2]={'5','+','A','B'}; // Invalid

In the above example, the compiler displays "initialize-string for array of chars is too long" error message.

Write a Program to check palindrome or not

```
#include<iostream>
#include<string.h>
using namespace std;
int main( )
{
        int i, j, len, flag =1;
        char a [20];
        cout<<"Enter a string:";
        cin>>a;
        for(len=strlen(a);a[len]!='\0';++len)
        for(i=0,j=len-1;i<len/2;++i,--j)
        {
        if(a[j]!=a[i])
                        flag=0;
                        break;
        }
```


11th Std Computer Science_EM Chapter 12.indd 213 06-01-2023 15:10:41